Contact

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Top Skills

Swagger API
SOAP
REST APIs

Languages

English (Full Professional)
German (Professional Working)

Certifications

Six Sigma Green Belt
REST API Design, Development and Management
Git
Generative AI in a Weekend: Learn ChatGPT and OpenAI
Linux

Honors-Awards

Indian School of Mines - Gold Medal
Kaushalya Devi Gold Medal

Harsh Sharma

IT Solutions for CAE - SPDM | SCALE India | DYNAmore GmbH | Stuttgart University (COMMAS) | Caterpillar | IIT(ISM) Dhanbad
Jaipur, Rajasthan, India

Summary

Managing SCALEsdm India Pvt. Ltd., an indian subsidiary of SCALE GmbH, a technology company based out of Germany, building cutting edge software solutions for CAE Simulation Process and Data Management (SPDM).

Overlooking company's operations in India with diverse activities, including hiring/team building, technical support and training for customers in India and Europe, business development, establishing company's footprint in India and assisting in company's day-to-day operations.

Passionate about computational mechanics and numerical methods for solving complex structural mechanics problems. A self driven, independent and results oriented simulation engineer with 8+ Years of computational modelling experience using FEM, including a Masters Degree in Computational Mechanics. Possess in-depth understanding of FE Solver LS-DYNA, including experience of customer support, consulting and training for LS-DYNA in general and topics related to car crash/occupant safety analysis in particular.

Experience

SCALE
Executive Director
July 2022 - Present (4 years 4 months)
Jaipur

DYNAmore
Simulation Engineer
April 2016 - June 2022 (6 years 3 months)
Stuttgart Area, Germany

University of Stuttgart

Graduate Student
October 2013 - March 2016 (2 years 6 months)

MSc. in Computational Mechanics of Materials and Structures (COMMAS), Grade 1.3

Hilti
Master Thesis Student
June 2015 - November 2015 (6 months)
Schaan

Daimler AG
Student Technical Assistant
June 2014 - June 2015 (1 year 1 month)

Caterpillar Inc.
2 years 3 months

Associate Engineer - Machine Structural Analysis
January 2013 - September 2013 (9 months)

My job role as a Machine Structural Analyst is to perform virtual validation of CAT machines using FE Analysis and Simulation techniques. I have worked on Domains of Linear/Non-Linear Static and Dynamic Analysis using commercially available as well as Caterpillar Internally developed Analysis tools. Through my projects, I have addressed the Issues of Field Failure, New Product Introduction as well as tool validation. Following is a list of key projects:

1. Transient Response Analysis of CAT 966L Large Wheel Loader Fender assembly to study the effects of adding a Fire Suppression system on Life numbers at critical locations, and alternative designs with improved life numbers.

2.Non Linear Static Analysis of CAT 988H Large Wheel Loader Engine Frame welding Fixture to evaluate the strength of Weld fixture, when the Frame is mounted on it under Gravity loading.

3.Non-Linear Static analysis of CAT 775G Truck Suspension Strut Cylinder to evaluate the weld throat crack initiation life when subjected to a series of load events: Material Interference between various contact surfaces, Bolt Pretension loads, and internal fluid pressure.

4.Forced Transient Response Analysis of CAT 745C Articulated Truck Fender assembly for addressing field failure issues for Proto1 build and proposing better designs with increased life numbers for subsequent builds.

5.Non-Linear Static analysis of 315D Cylinder Rod when subjected to Push-Pull loads and evaluate the Weld Throat crack initiation life using Battelle Structural Stress Method.

Leadership and Technical Development Program
July 2011 - December 2012 (1 year 6 months)

Leadership and Technical Development Program (LTDP) at Caterpillar exposed me to a multidisciplinary work experience in Earth Moving/ Mining equipment Industrial sector. It involved two rotations of 6 months at Engineering Design Center and 12 months at Manufacturing plant, respectively. This program gave me a rich experience of performing virtual validation of CAT machines using FEA, further, dealing with production issues of developing new Suppliers, addressing field failure issues,and cost reduction projects.

• Structural Simulation Projects – EDCI Machine Structural Analysis Team
1. Lifting Analysis of CAT Motor-Grader 12H to study the regions where stresses have exceeded safety limits.
2. G-Load and Modal Analysis of a Fuel Tank to study the sloshing effect.

• Production Projects – Caterpillar India Private Limited, Tiruvallur
1. Design Calculations for Mechanical Anti-Collision device for Caterpillar Off-Highway Truck
2. Development of SKF as a Bearing supplier for Caterpillar India Private Limited

TU Darmstadt
Research Assistant
June 2010 - August 2010 (3 months)

Worked as a Student Research Assistant at Institute of Systems Reliability and Machine Acoustics (SzM) in TU Darmstadt Germany, to study the contribution of in-plane waves propagating in a cantilever plate structure when excited with a harmonic force. A numerical formulation in MATLAB was developed to compute the In-Plane shear waves propagating through a vibrating thin plate structure. The output results were normalized shear wave intensity plots, which were compared with simulation results obtained from ANSYS and arrived at 80% confidence level. Practical implication of study is manipulating the energy flow through a structure to alter its dynamic properties.

Education

Universität Stuttgart

Master of Science (MS), Computational Mechanics · (2013 - 2015)

Indian Institute of Technology (Indian School of Mines), Dhanbad

B.Tech, Mechanical Engineering · (2007 - 2011)